AQI PM2.5 Demo

Show raw tables

Prediction rows

1000

AQI Big Data (PM2.5) Dashboard

Spark processing, EDA trends, and model predictions

Model Metrics

	model	rmse	mae	r2
0	linear_regression	3.4354	1.1873	0.9605

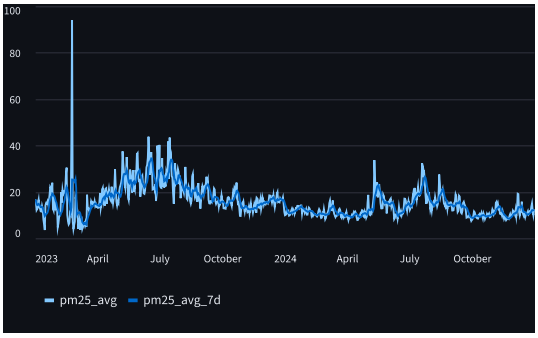
Best Model: linear_regression

R2: 0.9605

RMSE: 3.44

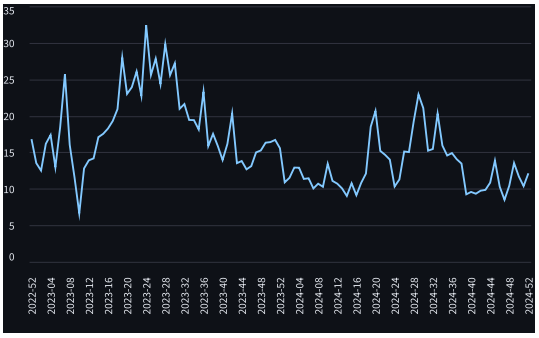
EDA Trends

Daily Trend (with 7-day moving average)



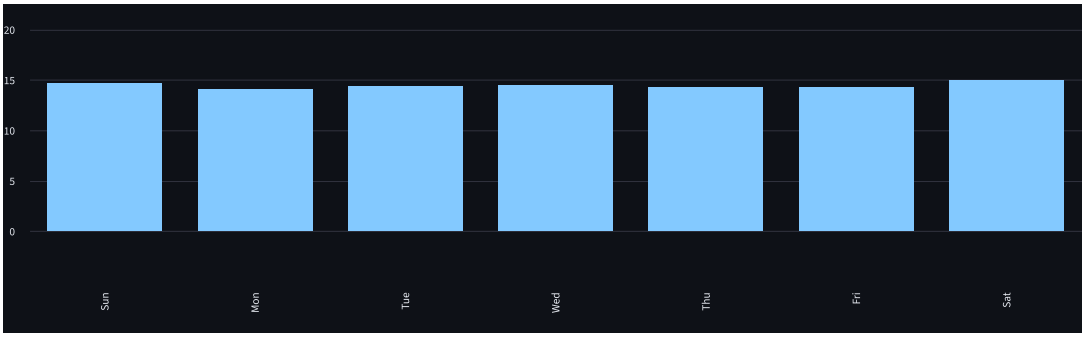
	date	pm25_avg	n	pm25_avg_7d
0	2022-12-31 00:00:00	16.8099	276	16.8099
1	2023-01-01 00:00:00	15.5845	857	16.1972
2	2023-01-02 00:00:00	12.1845	826	14.8596
3	2023-01-03 00:00:00	12.7102	868	14.3223
4	2023-01-04 00:00:00	15.144	868	14.4866
5	2023-01-05 00:00:00	16.699	887	14.8554
6	2023-01-06 00:00:00	13.2181	867	14.6215
7	2023-01-07 00:00:00	12.8227	832	14.0519
8	2023-01-08 00:00:00	11.6641	829	13.4918
9	2023-01-09 00:00:00	11.5807	875	13.4055

Weekly Trend



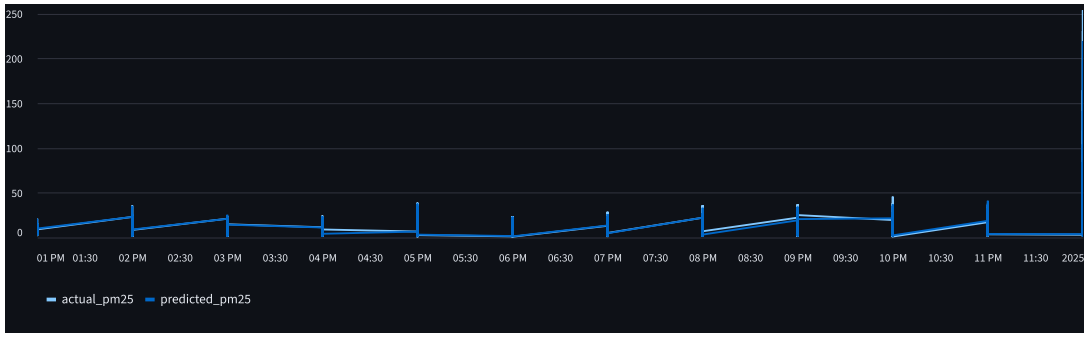
	year	week_of_year	pm25_avg	n	year_week
8	2023	8	16.0357	263	2023-08
9	2023	9	11.5907	237	2023-09
10	2023	10	6.6506	254	2023-10
11	2023	11	12.7979	1357	2023-11
12	2023	12	13.9081	4774	2023-12
13	2023	13	14.1715	5304	2023-13
14	2023	14	17.0827	5404	2023-14
15	2023	15	17.5332	4966	2023-15
16	2023	16	18.2548	4429	2023-16
17	2023	17	19.2896	4182	2023-17

Weekday Statistics



	day_of_week	n	pm25_avg	pm25_median	pm25_min	pm25_max	day_label
0	1	145586	14.6237	9	0.001	896	Sun
1	2	144321	14.0723	8.8	0.0074	896	Mon
2	3	143671	14.3628	8.916	0.0005	985	Tue
3	4	140245	14.4746	9	0.0005	580.1	Wed
4	5	138352	14.2597	9.0136	0.0025	1001.8	Thu
5	6	141677	14.2337	9	0.0015	985	Fri
6	7	143716	14.9336	9	0.0015	1323.2	Sat

Predictions (Actual vs Predicted)



	date	actual_pm25	predicted_pm25
352636	2024-12-31 13:00:00	15	14.9276
555368	2024-12-31 13:00:00	1.6	2.7759
68372	2024-12-31 13:00:00	10.8	11.0087
254067	2024-12-31 13:00:00	11.3	8.9481
177355	2024-12-31 13:00:00	8.2472	8.2745
656821	2024-12-31 13:00:00	9.533	10.2368
415936	2024-12-31 13:00:00	3.3	3.3227
374444	2024-12-31 13:00:00	3.1	2.8278
203008	2024-12-31 13:00:00	13.1	13.7907
473343	2024-12-31 13:00:00	11.3	11.0933

Tip: Set the Spark's to refresh auto-int before demos.